ITUC201: Fundamentals of computer networks
Name: Prince Sakariya **25DIT073**

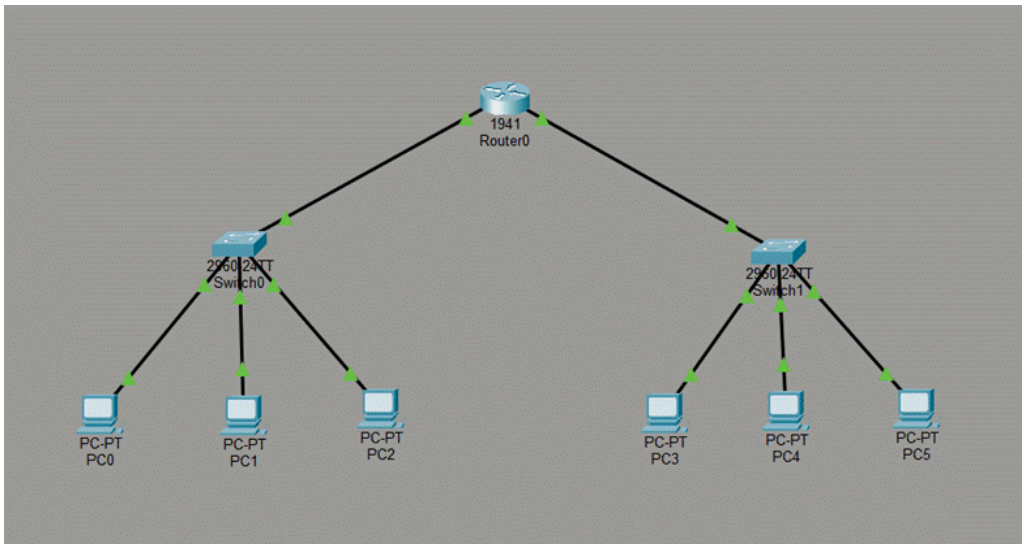
Practical 3

Problem Definition:

Problem Definition: Configure two independent LANs connected through a router and verify communication within and across networks.

Device	IP Address	Subnet Mask	Gateway
PC0	192.168.1.10	255.255.255.0	192.168.1.1
PC1	192.168.1.20	255.255.255.0	192.168.1.1
PC2	192.168.1.30	255.255.255.0	192.168.1.1
Router G0/0	192.168.1.1	255.255.255.0	-
PC3	192.168.2.10	255.255.255.0	192.168.2.1
PC4	192.168.2.20	255.255.255.0	192.168.2.1
PC5	192.168.2.30	255.255.255.0	192.168.2.1
Router G0/1	192.168.2.1	255.255.255.0	-

Solution:



OUTPUT :

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Different lane:

```
C:\>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Reply from 192.168.1.10: bytes=32 time<1ms TTL=127
Reply from 192.168.1.10: bytes=32 time<1ms TTL=127
Reply from 192.168.1.10: bytes=32 time<1ms TTL=127
Reply from 192.168.1.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
C:\>ping 192.168.1.20

Pinging 192.168.1.20 with 32 bytes of data:

Reply from 192.168.1.20: bytes=32 time<1ms TTL=127
Reply from 192.168.1.20: bytes=32 time=1ms TTL=127
Reply from 192.168.1.20: bytes=32 time<1ms TTL=127
Reply from 192.168.1.20: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.1.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

```
C:\>ping 192.168.1.30

Pinging 192.168.1.30 with 32 bytes of data:

Reply from 192.168.1.30: bytes=32 time<1ms TTL=127
Reply from 192.168.1.30: bytes=32 time<1ms TTL=127
Reply from 192.168.1.30: bytes=32 time<1ms TTL=127
Reply from 192.168.1.30: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.1.30:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

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```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.20

Pinging 192.168.1.20 with 32 bytes of data:

Reply from 192.168.1.20: bytes=32 time=6ms TTL=128
Reply from 192.168.1.20: bytes=32 time=3ms TTL=128
Reply from 192.168.1.20: bytes=32 time=17ms TTL=128
Reply from 192.168.1.20: bytes=32 time=4ms TTL=128

Ping statistics for 192.168.1.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 17ms, Average = 7ms

C:\>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time=1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time=2ms TTL=128

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>ping 192.168.1.30

Pinging 192.168.1.30 with 32 bytes of data:

Reply from 192.168.1.30: bytes=32 time<1ms TTL=128
Reply from 192.168.1.30: bytes=32 time=1ms TTL=128
Reply from 192.168.1.30: bytes=32 time<1ms TTL=128
Reply from 192.168.1.30: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.30:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

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Same lane:

```
C:\>ping 192.168.2.20

Pinging 192.168.2.20 with 32 bytes of data:

Reply from 192.168.2.20: bytes=32 time<1ms TTL=128
Reply from 192.168.2.20: bytes=32 time<1ms TTL=128
Reply from 192.168.2.20: bytes=32 time<1ms TTL=128
Reply from 192.168.2.20: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.2.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.2.30

Pinging 192.168.2.30 with 32 bytes of data:

Reply from 192.168.2.30: bytes=32 time<1ms TTL=128
Reply from 192.168.2.30: bytes=32 time<1ms TTL=128
Reply from 192.168.2.30: bytes=32 time=1ms TTL=128
Reply from 192.168.2.30: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.2.30:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.20

Pinging 192.168.1.20 with 32 bytes of data:

Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.30

Pinging 192.168.1.30 with 32 bytes of data:

Reply from 192.168.1.30: bytes=32 time<1ms TTL=128
Reply from 192.168.1.30: bytes=32 time<1ms TTL=128
Reply from 192.168.1.30: bytes=32 time<1ms TTL=128
Reply from 192.168.1.30: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.30:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

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Learning:

→ Learned to configure two LANs using a router by assigning correct IP addresses, subnet masks, and default gateways. Successfully verified communication within the same network and between different networks using the **ping** command.